

Assessment of a coupled finite-volume framework for modelling moored floating solar platforms

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1. Tabulated Results

Surface Elevation		Experiment FFT Amp	Model FFT Amp	Difference	% Difference
Frequency (Hz)	Amplitude (cm)				
0.3	3	0.048 m	0.048 m	0.000 m	0.26
0.3	6	0.085 m	0.088 m	0.003 m	3.57
0.5	3	0.036 m	0.036 m	0.000 m	0.33
0.5	6	0.071 m	0.071 m	0.000 m	0.25
0.5	9	0.103 m	0.103 m	0.000 m	0.09
0.7	3	0.033 m	0.034 m	0.001 m	1.51
0.7	6	0.065 m	0.061 m	−0.004 m	7.06
0.7	9	0.098 m	0.091 m	−0.007 m	6.90
0.9	3	0.030 m	0.029 m	−0.001 m	5.26
0.9	6	0.060 m	0.061 m	0.001 m	1.71
0.9	9	0.086 m	0.092 m	0.006 m	7.56
1.1	3	0.037 m	0.035 m	−0.002 m	3.77
1.1	6	0.067 m	0.068 m	0.001 m	0.72

Table 1: Surface Elevation FFT results

Pitch		Experiment FFT Amp	Model FFT Amp	Difference	% Difference
Frequency (Hz)	Amplitude (cm)				
0.3	3	2.414°	6.668°	4.254°	176.21
0.3	6	6.725°	12.379°	5.654°	84.07
0.5	3	7.146°	6.280°	−0.866°	12.12
0.5	6	12.133°	10.546°	−1.587°	13.08
0.5	9	15.110°	13.684°	−1.426°	9.44
0.7	3	5.255°	4.896°	−0.359°	6.82
0.7	6	8.903°	9.005°	0.102°	1.16
0.7	9	12.869°	13.252°	0.383°	2.98
0.9	3	3.743°	5.681°	1.938°	51.79
0.9	6	8.182°	11.840°	3.658°	44.70
0.9	9	11.873°	17.200°	5.327°	44.87
1.1	3	4.236°	12.145°	7.909°	186.69
1.1	6	10.857°	17.501°	6.644°	61.19

Table 2: Pitch FFT results

Surge		Experiment FFT Amp	Model FFT Amp	Difference	% Difference
Frequency (Hz)	Amplitude (cm)				
0.3	3	0.036 m	0.105 m	0.069 m	189.97
0.3	6	0.104 m	0.186 m	0.082 m	78.92
0.5	3	0.088 m	0.074 m	−0.014 m	15.92
0.5	6	0.157 m	0.122 m	−0.035 m	22.39
0.5	9	0.193 m	0.154 m	−0.039 m	20.11
0.7	3	0.032 m	0.035 m	0.003 m	8.92
0.7	6	0.061 m	0.062 m	0.001 m	2.50
0.7	9	0.089 m	0.088 m	−0.001 m	1.36
0.9	3	0.012 m	0.013 m	0.001 m	3.87
0.9	6	0.024 m	0.028 m	0.004 m	14.35
0.9	9	0.036 m	0.044 m	0.008 m	23.73
1.1	3	0.007 m	0.017 m	0.010 m	159.88
1.1	6	0.017 m	0.035 m	0.018 m	102.31

Table 3: Surge FFT results

Heave		Experiment FFT Amp	Model FFT Amp	Difference	% Difference
Frequency (Hz)	Amplitude (cm)				
0.3	3	0.047 m	0.047 m	0.000 m	0.74
0.3	6	0.085 m	0.088 m	0.003 m	3.83
0.5	3	0.038 m	0.036 m	-0.002 m	4.74
0.5	6	0.075 m	0.072 m	-0.003 m	2.98
0.5	9	0.105 m	0.107 m	0.002 m	1.96
0.7	3	0.042 m	0.035 m	-0.007 m	15.92
0.7	6	0.084 m	0.065 m	-0.019 m	22.46
0.7	9	0.126 m	0.104 m	-0.022 m	17.11
0.9	3	0.075 m	0.068 m	-0.007 m	9.71
0.9	6	0.118 m	0.100 m	-0.018 m	15.39
0.9	9	0.132 m	0.118 m	-0.014 m	11.30
1.1	3	0.014 m	0.010 m	-0.004 m	29.54
1.1	6	0.017 m	0.014 m	-0.003 m	17.93

Table 4: Heave FFT results

Front Tension		Experiment FFT Amp	Model FFT Amp	Difference	% Difference
Frequency (Hz)	Amplitude (cm)				
0.3	3	1.516 N	4.582 N	3.066 N	202.22
0.3	6	4.521 N	8.048 N	3.527 N	78.02
0.5	3	3.461 N	2.840 N	-0.621 N	17.95
0.5	6	6.301 N	4.687 N	-1.614 N	25.63
0.5	9	7.729 N	5.970 N	-1.759 N	22.75
0.7	3	0.744 N	0.859 N	0.115 N	15.47
0.7	6	1.599 N	1.581 N	-0.018 N	1.12
0.7	9	2.420 N	2.372 N	-0.048 N	1.98
0.9	3	0.134 N	0.669 N	0.535 N	399.74
0.9	6	0.454 N	1.440 N	0.986 N	217.42
0.9	9	0.748 N	2.164 N	1.416 N	189.23
1.1	3	0.602 N	3.391 N	2.789 N	463.58
1.1	6	2.352 N	4.727 N	2.375 N	100.98

Table 5: Front Tension FFT results

Rear Tension		Experiment FFT Amp	Model FFT Amp	Difference	% Difference
Frequency (Hz)	Amplitude (cm)				
0.3	3	1.464 N	4.640 N	3.176 N	216.91
0.3	6	4.150 N	7.912 N	3.762 N	90.64
0.5	3	3.385 N	2.877 N	-0.508 N	15.01
0.5	6	6.126 N	4.699 N	-1.427 N	23.28
0.5	9	7.381 N	5.811 N	-1.570 N	21.28
0.7	3	0.723 N	0.869 N	0.146 N	20.27
0.7	6	1.520 N	1.522 N	0.002 N	0.18
0.7	9	2.222 N	1.966 N	-0.256 N	11.48
0.9	3	0.101 N	0.681 N	0.580 N	573.33
0.9	6	0.480 N	1.585 N	1.105 N	230.47
0.9	9	0.807 N	2.588 N	1.781 N	220.68
1.1	3	0.584 N	3.438 N	2.854 N	488.71
1.1	6	2.223 N	5.059 N	2.836 N	127.56

Table 6: Rear Tension FFT results